

Final Collection CPIT220

MAFIA: <https://t.me/FCIT20Male>

تجميعية للاختبار تقريباً من 60-70 سؤال.

أيش الغرض من التيسر بانك؟ تعرف طبيعة الاختبار وتتوقع ايش
بيجيك، ماهو مصدر تعتمد عليه 100%، لكن توقع يجيك شيء
مشابه مثله .. بالتوفيق يا حلوين ورمضان مبارك.

الفهرس:

- نموذج 1: نموذج فاينل حقيقي، مصدره قروبات الواتس..
- نموذج 2: مصدره درايف 19، بنت حلتها + بحط لمساتي
ونفشحه هع..
- نموذج 3: كويز عن Data Base من د. حسام، شكراً F
حليته ..
- نموذج 4: كويز عن شبتر System analysis & design
من د. حسام، شكراً م. م. حليته ..
- نموذج 5: حلول فاينل العملي عن Data Base، شكراً ابو
زهرة لحلك ..
- نموذج 6: كويزات من دفعة 18 عن شبتر Data Base +
.. Audio

نموذج 1

CLO 5 - SO (a)

3
3

Question # 1.

- a. Digitization of audio signal is achieved in three steps. List these steps along with their description. (1.5 marks)

- Sampling: ~~convert the analogue~~ to digital.
- Quantizing: ~~rounding off~~ voltage to nearest value
- Coding: Assigned a binary code for each sample

- b. Calculate the total number of bits for an audio signal with a bandwidth of 4000 Hz that has lasted for 4 minutes if the number of bits in every sample is 8 bits? (1.5 marks)

~~No~~ of samples: $T \times f_s$

$$= (4 \times 60) \times 2(4000) = 1920000$$

~~No~~ of bits = 1920000×8

$$= 15360000$$

115 MB

CLO 6-S0(a)

3

Question # 2.

a. What is Bitmap image used for? Write a formula to calculate total number of bits in the file? (1 mark)

used to store images in files.

$$\boxed{N * S}$$

↓ ↓
of bits size of image

b. Calculate the number of megabytes in a color image captured by a digital camera that can produce images with 1600x1200 pixels with bit depth of 24? How many images can be stored in the camera's 64 GB flash memory? (2 marks)

$$\# \text{ of Pixels} = 1600 * 1200 = 1920000$$

$$\# \text{ of bits} = 1920000 * 24 = 46080000$$

$$\# \text{ of bytes} = 46080000 / 8 = 5760000$$

$$\# \text{ of Megabytes} = 5760000 / 2^{20}$$

$$= 5.49316$$

5.76 MB

0.55 526 GB

HVCLO 7- SO(j)

Question # 3.

Answer the following questions.

a. Consider Figure 1 and write SQL query to insert a record. (1 mark)

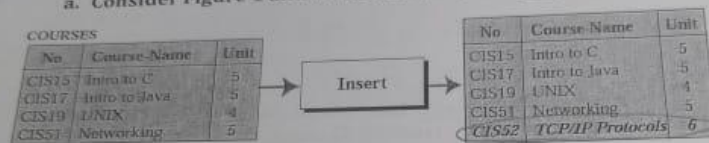


Figure 1

insert into course-Name values('CIS52', 'TCP/IP Protocols', 6);

b. Consider Figure 2 and write SQL query of Join operation. (1 mark)

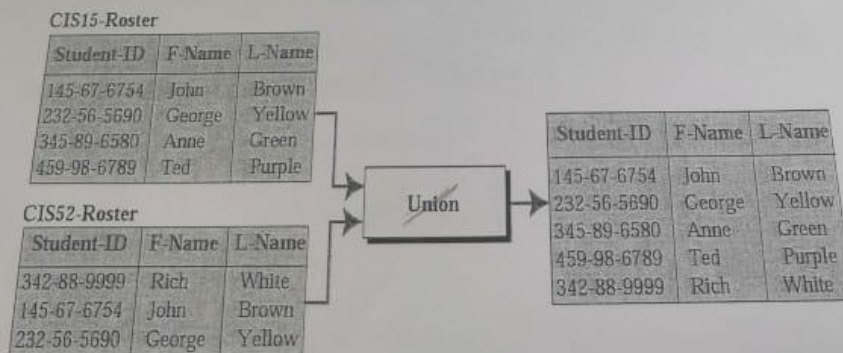


Figure 2

select * from CIS15-Roster Join select * from CIS52-Roster

c. What is data mart concept in a data warehouse? (1 mark)

data warehouse: is combining data or organizing all data in one large database

data mart is smaller or data warehouse but dealing with department or users

d. Consider the Table below and answer the following terms of a relation in the database. (2.25 marks)

COURSES			
Course#	Title	CrHour	InstName
DIS 110	Introduction to DOS	2	Lujan
DIS 118	Microcomputer Applications	3	Wilke
DIS 138	Introduction to Windows	2	Lujan
DIS 140	Introduction to Database/Access	3	Presley
DIS 150	Introduction to Spreadsheet/Excel	2	Morris

i. Mention the number of attributes.

4

ii. List the attributes of 5th tuple? There is 1st tuple:

tuple: collection of attributes
course#, Title, CrHour, InstName

iii. What is the cardinality?

collection of rows

5 rows

iv. What is the degree?

collection of columns

4 degree

v. Mention the name of above relation.

Relational Database Systems

COURSES

RDBS

CLO 8- SO(m)

Question # 4.

2.5
3.75

a. Write down the deliverables of the following phases: (1.5 marks)

Phases	Deliverables
Phase-1 (Planning)	Made a plan for process containing all levels
Phase-2 (Analyzing and Documenting the Existing System)	Analyze data and collect it data.
Phase-3 (System Design)	after analyzing you design it through many steps.

b. List any three methods of system conversion. (1.5 marks)

No.	Conversion method
1	Parallel conversion
2	Pilot conversion
3	Direct conversion

c. Write down the deliverable of phase 5. (0.75 mark)

Plan
Analyze
Design
Implementation
Maintenance

Maintain Phase
Solve any problem that will happen

0.75

نموذج 2

(موسرط تمشي بنفس حلها لأنها هي فصلت بزيادة وحطت شرح،
لكن ركز على السؤال وكون اجابة مشابهه مناسبة لك اذا ما
عجبتك اجابتها اللي من الكتاب ..)

1

1. For any electrical signal. Define the following terms.

Chapter 10 Fundamentals of Communications

- a. Amplitude? **The maximum value of a signal, measured from its average state**
- b. Frequency? **The number of cycles produced in a second – Hertz (Hz).**

2

2. Modulation is a technique used to convert a low frequency signal (information signal) to a higher frequency signal (modulated signal) using a higher frequency signal (carrier).

Chapter 10 Fundamentals of Communications

- a. What types of modulations are there?

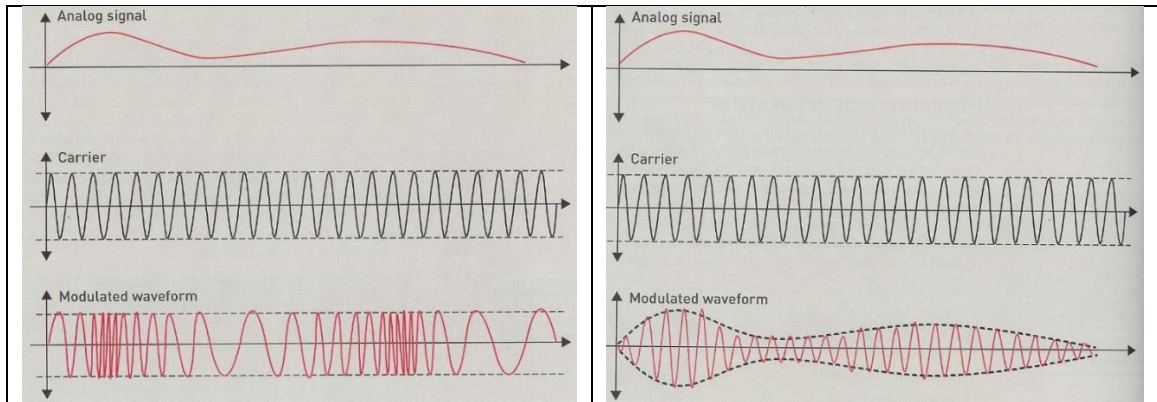
Analogue Modulation:

- **The main idea of analogue modulation is to carry the information signal onto the carrier's amplitude, frequency or phase.**

digital modulation:

- **Digital information is made of 2 levels of amplitude representing two logical levels (0 and 1)**
- **In digital modulation the 0 and 1 are carried on the carrier's amplitude, frequency and phase**

b. Identify the following modulation techniques.



Frequency Modulation

Amplitude Modulation

2

3. Error control coding (ECC) is the name of techniques used to encode the message prior to transmission. Also, Block coding is a kind of ECC and consists of two type. What are they?

Chapter 10 Fundamentals of Communications

- a. **Single Parity Checking:** Single Parity Checking is implemented simply by adding a single redundant bit called the parity bit
- b. **Rectangular Coding:** Rectangular Coding is more complex than single parity checking.
These coding technique can detect and correct errors.

4. What are the categories of computer networks?

Chapter 11 Network.

1-Local area network (LAN)

- (Uses cables, radio waves, or infrared signals

Links computers in a limited geographic area)

هذا الكلام اللي تحته خط مو مره مهم بس يعني اقرأوه لأنه التفاصيل حلوه تبين الفرق، هي ٨٨٨ البنت معرقه شوي + السؤال طلب كاتيجوري يعني أنواع، لكن ما طلب شرح للأنواع ..

2-Wide area network (WAN)

Uses long-distance transmission media

Links computer systems a few miles or thousands of miles

Internet is the largest WAN

نفس الهرجة ٨٨٨

3-Metropolitan area network (MAN)

Designed for a city

Larger than a LAN, smaller than a WAN

نفس الهرجة ٨٨٨

4-Campus area network (CAN)

Several LANs located in various locations on a college or business campus

Smaller than a WAN

Use devices such as switches, hubs, and routers

٨٨٨ نفس الهرجة

5-Personal area network (PAN)

Network of an individual's own personal devices

Usually use wireless technology .. زي البلوتوث يعني ..

نفس الهرجة ٨٨٨

1

5. List advantages and disadvantages of computer networks?

Chapter 11 Network.

Advantages

Reduced hardware costs

Application sharing

Sharing information resources

Data management centralization

Connecting people

Disadvantages

Loss of autonomy

Lack of privacy

Security threats

Loss of productivity

1

6. For local area networks, list three different topologies along with their descriptions and draw how they look like.

Chapter 11 Network.

(السؤال طلب رسمه، لكن البنت تتغلى ترسم، يمكن مافحيلها، هع.. المهم يا عيال ارسموا يهوه حتى انا
مفحيلي ارسوم)

1-Bus topology

Practical for home or small office

One node transmits at a time

Terminators signify the end of the circuit

Uses contention management—technique that specifies what happens
when a collision occurs

2-Star topology

For office buildings, computer labs, and WANs

Easy to add users

3-Ring topology

For a division of a company or one floor

Not in common use today

Node can transmit only when it has the token—special unit of data that travels around the ring

2

7. What are the differences between circuit switching and packet switching?

Chapter 11 Network.

Circuit switching:

Used by the public switched telephone network

to send data over a physical end-to-end circuit

Provides a direct connection between devices

Packet switching:

Used for computer communication

Divides and sends outgoing messages as packets, which are reassembled on receipt

More efficient and less expensive than circuit switching

1

8. Define privacy and anonymity.

Chapter 12 Privacy, Crime, and Security.

Privacy:

Individual's ability to restrict or eliminate the collection, use, and sale of confidential personal information

Anonymity:

Means to communicate without disclosing one's identity

More difficult with the use of computers and the Internet

2

9. What is the difference between Cookies and Spam?

Chapter 12 Privacy, Crime, and Security.

Cookies:

Small files written to your hard disk by Web sites visited

Examples include:

Track your browsing habits

Gather personal information without your consent

Can be disabled

SPAM:

Unsolicited messages sent in bulk over electronic mailing systems

CAN-SPAM Act of 2003

U.S. law that provided tools to combat spammers.

2

10. Differentiate between Malware and Spyware.

Chapter12 Privacy, Crime, and Security.

Malware (short for malicious software): programs that intentionally harm a computer system or allow individuals to gain access without permission

Spyware: software that gathers private information and tracks Web use

1

11. What is cryptography?

Chapter12 Privacy, Crime, and Security.

Study of transforming information into an encoded or scrambled format

1

12. Describe plaintext, encryption key, and ciphertext?

Chapter12 Privacy, Crime, and Security.

Plaintext: Readable message that has not been encrypted

Encryption key: Formula that makes a plaintext message unreadable

Ciphertext: Coded message

2

13. Differentiate between symmetric and asymmetric cryptography.

Chapter12 Privacy, Crime, and Security.

Symmetric key encryption:

Uses same key for both encryption and decryption

Public key encryption:

Also referred to as **asymmetric key encryption**

Uses two keys:

Public key to encrypt

Private key to decrypt

Essential for e-commerce

Used to implement:

Digital signatures—guarantee messages are secure

Digital certificates—validate identity

14. مسألة حسابية ب :

Chapter 10 Fundamentals of Communications

المسألة وشرحها

Slides → 42,43, 44, 45, ..

100110
↪ 1001101

"عشان اتكرر رقم 1
3 مرات وهو عدد فردي
فنزيد 1"

1100101
↪ 11001010

"عشان اتكرر رقم 1
4 مرات وهو عدد زوجي
فنزيد 0"

نموذج 3

Question 1

A ----- is a field that is used in an RDBMS table as an index to locate records.

- ☐ record key
- ☐ table key
- ☒ primary key
- ☐ foreign key

Question 2

Compared to file management systems, RDBMS is considered:

- ☐ Less expensive.
- ☐ bad in controlling data redundancy.
- ☐ All of the above
- ☒ None of the above

Question 3

0.5 points

----- is a collection of related data organized so it is easy to access, manage, update, group, summarize

- ☐ 1. Records
- ☐ 2. Fields
- ☒ 3. Database
- ☐ 4. Tables

Question 4

0.5 poi

A ----- includes a group of related fields contained in tables or data files, and it has a primary key.

- ☒ 1. Record
- ☐ 2. Field
- ☐ 3. Data
- ☐ 4. System

Question 5

The different classifications of DBMS are:

- ☒ Object Oriented
- ☐ Utilities Program
- ☐ Kernel
- ☒ Multidimensional
- ☒ Relational
- ☐ SQL

Question 6

The following describes multidimensional databases (check all that applies)

- ☐ 1. Multidimensional databases manage flat files
- ☐ 2. Multidimensional databases manage multimedia files
- ☒ 3. is represented as data cubes
- ☒ 4. dimensions represents categories
- ☒ 5. use MDX query language
- ☐ 6. data stored in tables

Question 7

----- is an SQL operation that creates relation in which each tuple has fewer attributes.

- ☒ project operation
- ☐ select operation
- ☐ update
- ☐ delete

Question 8

----- is an SQL operation that changes the values of some attributes of a tuple.

- ☐ 1. project operation
- ☒ 2. update operation
- ☐ 3. insert operation
- ☐ 4. delete operation

Question 9

0.5 points ✓ Saved

The difference between File Management Program and DBMS is that File Management program generates a ----- and DBMS manage -----.

- ☐ 1. records, fields
- ☒ 2. Flat file, multiple tables or files.
- ☐ 3. rows, columns
- ☐ 4. software, hardware

نموذج 4

Question 1

1 points

✓ Saved

----- is the person responsible for determining information system requirements, working with users and management, and having good listening and communication skills

- ☐ A. Information System Steering Committee
- ☐ B. Project Manager
- ☒ C. System Analyst
- ☐ D. Project Team

Q2/

----- provides structure for system development and seeks to improve system quality.

- ☒ A. SDLS
- ☐ B. SSH
- ☐ C. LSTL
- ☐ D. SSL

Q3/

----- involves information systems planning, development, and implementation of artificial systems.

- ☐ A. System Planning
- ☐ B. System Development
- ☒ C. System Analysis
- ☐ D. System Design

Q4/

one of the modern approaches in software development is ----- which is a small mock up of a system.

- ☒ A. prototyping
- ☐ B. JAD
- ☐ C. Agile
- ☐ D. RAD

Q5/

Systems development life cycle consists of five phases:

- ☐ A. planning, analysis, testing, quality assurance, delivery
- ☐ B. development, research, testing, quality assurance, delivery
- ☒ C. Planning, analysis, design, implementation, maintenance
- ☐ D. planning, development, testing, quality, delivery

نموذج 5

IT-courses

NO	Courses- Name	Unit
CPIT 220	Intro to IT	4
CPIT 265	Intro to Java	4
CPIT 280	Intro to Media	4
CPIT 330	Network	4

IT-Taught

Taught no	Taught Proof
CPIT 220	X
CPIT 265	Y
CPIT 280	Z
CPIT 330	W



IT-Student

Student ID	Student Fname	Student Lname
1900000	Ahmad	Naser
2000000	Naser	Ahmad
2100000	Sroor	Sari
2200000	Sari	Sroor



IT-Student1

Student ID	Student Fname	Student Lname
1800000	Mohammad	Mostafa
1900000	Ahmad	Naser
2200000	Sari	Sroor
2300000	Khaled	Ahmad

Q1/

Could you please create tables?

```
SQL> create table IT_Courses(courses_no varchar(10), courses_name varchar(20), courses_unit number(1));
Table created.

SQL> create table IT-Taught(taught_no varchar(10), taught_proff varchar(20));
Table created.

SQL> create table IT_Student(sudent_id number(15), student_FName varchar(10), student_LName varchar(10));
Table created.

SQL> create table IT_Student1(sudent_id number(15), student_FName varchar(10), student_LName varchar(10));
Table created.

SQL>
```

Q2/

Could you please insert rows?

IT_courses Table

```
SQL> insert into IT_courses values('CPIT 220', 'Intro to IT', 4);
1 row created.

SQL> insert into IT_courses values('CPIT 265', 'Intro to Java', 4);
1 row created.

SQL> insert into IT_courses values('CPIT 280', 'Intro to Media', 4);
1 row created.

SQL> insert into IT_courses values('CPIT 330', 'Network, 4);
ERROR:
ORA-01756: quoted string not properly terminated

SQL> insert into IT_courses values('CPIT 330', 'Network', 4);
1 row created.
```

IT_taught Table

```
SQL> insert into IT_taught values('CPIT 220', 'X');
1 row created.

SQL> insert into IT_taught values('CPIT 265', 'Y');
1 row created.

SQL> insert into IT_taught values('CPIT 280', 'Z');
1 row created.

SQL> insert into IT_taught values('CPIT 330', 'W');
1 row created.
```

IT_student Table

```
SQL> insert into IT_student values(1900000, 'Ahmad', 'Naser');
1 row created.

SQL> insert into IT_student values(2000000, 'Naser', 'Ahmad');
1 row created.

SQL> insert into IT_student values(2100000, 'Sroor', 'Sari');
1 row created.

SQL> insert into IT_student values(2200000, 'Sari', 'Sroor');
1 row created.
```

IT_student1 Table

```
SQL> insert into IT_student1 values(1800000, 'Mohammad', 'Mostafa');
1 row created.

SQL> insert into IT_student1 values(1900000, 'Ahmad', 'Naser');
1 row created.

SQL> insert into IT_student1 values(2200000, 'Sari', 'Sroor');
1 row created.

SQL> insert into IT_student1 values(2300000, 'Khaled', 'Ahmad');
1 row created.
```

Q3/

K A U

Could you please let me see tables' information?

```
select * from IT_courses;
```

```
SQL> select * from IT_courses;
```

COURSES_NO	COURSES_NAME	COURSES_UNIT
CPIT 220	Intro to IT	4
CPIT 265	Intro to Java	4
CPIT 280	Intro to Media	4
CPIT 330	Network	4

```
select * from IT_taught;
```

```
SQL> select * from IT_taught;
```

TAUGHT_NO	TAUGHT_PROFF
CPIT 220	X
CPIT 265	Y
CPIT 280	Z
CPIT 330	W

```
select * from IT_student
```

```
SQL> select * from IT_student;
```

SUDENT_ID	STUDENT_FN	STUDENT_LN
1900000	Ahmad	Naser
2000000	Naser	Ahmad
2100000	Sroor	Sari
2200000	Sari	Sroor

Select * from IT_student1

```
SQL> Select * from IT_student1;
```

SUDENT_ID	STUDENT_FN	STUDENT_LN
1800000	Mohammad	Mostafa
1900000	Ahmad	Naser
2200000	Sari	Sroor
2300000	Khaled	Ahmad

Q4/

Could you please Select from **IT- courses table** where IT-courses_NO = CPIT220?

select * from IT_courses where courses_no ='CPIT 220';

```
SQL> select * from IT_courses where courses_no ='CPIT 220';
```

COURSES_NO	COURSES_NAME	COURSES_UNIT
CPIT 220	Intro to IT	4

Q5/

Could please use **union** statement between two tables of students?

```
SQL> select * from IT_student union select * from IT_student1;
```

SUDENT_ID	STUDENT_FN	STUDENT_LN
1800000	Mohammad	Mostafa
1900000	Ahmad	Naser
2000000	Naser	Ahmad
2100000	Sroor	Sari
2200000	Sari	Sroor
2300000	Khaled	Ahmad

6 rows selected.

Q6/

Could you please choose **different** information between two tables of students?

```
SQL> select * from IT_student minus select * from IT_student1;
```

SUDENT_ID	STUDENT_FN	STUDENT_LN
2000000	Naser	Ahmad
2100000	Sroor	Sari

Q7/

Could you please **delete** one row from any tables of students?

```
SQL> delete from IT_student where student_FName ='Ahmad';
```

1 row deleted.

Q8/

Could you please show me table details after you delete row?

```
SQL> select * from IT_student;
```

SUDENT_ID	STUDENT_FN	STUDENT_LN
2000000	Naser	Ahmad
2100000	Sroor	Sari
2200000	Sari	Sroor

نموذج 6

Q1/

Question 1

DBMS is a collection of that enables a user to create and maintain a database.

- ☐ Keys
- ☐ Translators
- ☒ Programs
- ☐ Language Activity

Q2/

The minimum value of sampling rate (f_s) should be

- ☐ at least twice as much as the signal's bandwidth
- ☐ equal to the signal's bandwidth
- ☐ $f_s \geq 2 B$
- ☒ a and c

Q3/

COURSES

No	Course-Name	Unit
CIS15	Intro to C	5
CIS17	Intro to Java	5
CIS19	UNIX	4
CIS51	Networking	5
CIS52	TCP/IP Protocols	6

Update

No	Course-Name	Unit
CIS15	Intro to C	5
CIS17	Intro to Java	5
CIS19	UNIX	4
CIS51	Networking	6
CIS52	TCP/IP Protocols	6

Updating Query for the above diagram using SQL will be

- ☐ Set Courses Update Unit=6 where No="CIS51"
- ☒ Update Courses Set Unit=6 where No="CIS51"
- ☐ Update the Table automatically and set No= "CIS51"
- ☐ Set Courses Update Unit=5 where No="Networking"

Q4/

Which of the following steps are sequential steps of digitization process?

- ☒ Sampling, quantizing and coding
- ☐ Quantizing, coding and sampling
- ☐ Coding, quantizing and sampling
- ☐ None of the above

Q5/

Which of the following is an example of a DBMS?

- ☒ Microsoft Access
- ☐ Excel
- ☐ Word 2007
- ☐ None of the above

Q6/

Which one of the following operations creates a new relation in which tuple is either in both relations?

- ☐ Union
- ☒ Intersection
- ☐ Difference
- ☐ Join

Q7/

How long would it take to download a 4-minute audio over a 56k bps? Assuming that the file is sampled at 44.1k Hz using 16bit ADC? (Hint: # of sample, # of bits)

- ☐ 33.5 min
- ☐ 31.5 min
- ☐ 39.4 min
- ☒ 50.4 min

Q8/

Calculate the total number of bits for an audio signal with a bandwidth of 4000 Hz that has lasted for 4 minutes if the number of bits in every sample is 8 bits. (Hint: $t = T * fs$)

- ☒ 15360 k
- ☐ 2880 k
- ☐ 23040 k
- ☐ 13360 k

Q9/

An advantage of DBMS is to avoid the repetition of the entered data, this characteristic is known as..... ?

- ☐ data validation
- ☐ drill down technique using data mining
- ☐ data inspection
- ☒ avoiding data redundancy

Q10/

Which of the following statements is true about ODBMS (Object-oriented database management systems).

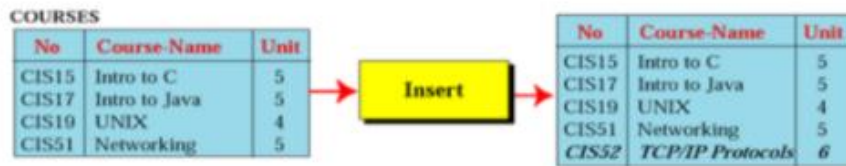
- ☐ Suits media applications
- ☐ Data represented as objects
- ☐ Incorporate sound, video, text, and graphics
- ☒ All of the above

Q11/

The speaker is an electric device that is used to convert

- ☒ An electrical signal into the sound waves
- ☐ Mechanical sound into electrical signal
- ☐ Analog to digital signal
- ☐ None of the above

Q12/



Inserting Query for the above diagram using SQL will be

- insert into Courses values("CIS51", "TCP/IP Protocols", 5)
- add into Courses("CIS52", "TCP/IP Protocols", 6)
- Plus into courses values("CIS52", "TCP/IP Protocols", 6)
- insert into Courses values("CIS52", "TCP/IP Protocols", 6)

کفو یا فحل خلصت، بالتوفیق بالاختبار